

LABORATORY OF AUTOMATION SYSTEMS

Davide Samorì

Dipartimento di Ingegneria dell'Energia Elettrica e dell'Informazione (DEI)

Università di Bologna

Email: davide.samori@unibo.it

SPEED/POSITION

In our setup, to detect speed and position, we have an encoder.

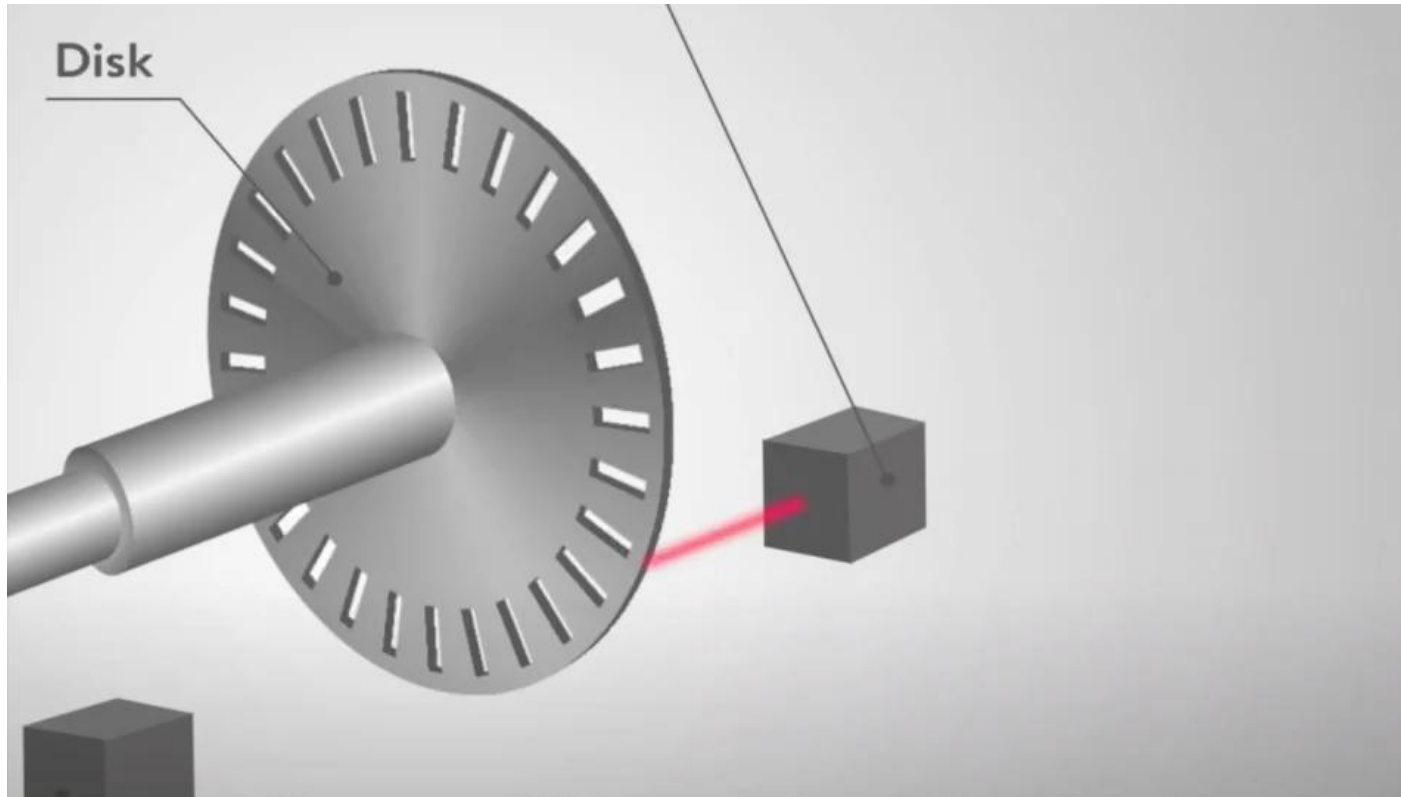


Inside an encoder we have:

- A shaft connected to a disk
- A disk with n slot (n = resolution)
- A light emitter
- A light receiver
- An electronic circuit (to convert light signal into a square wave)

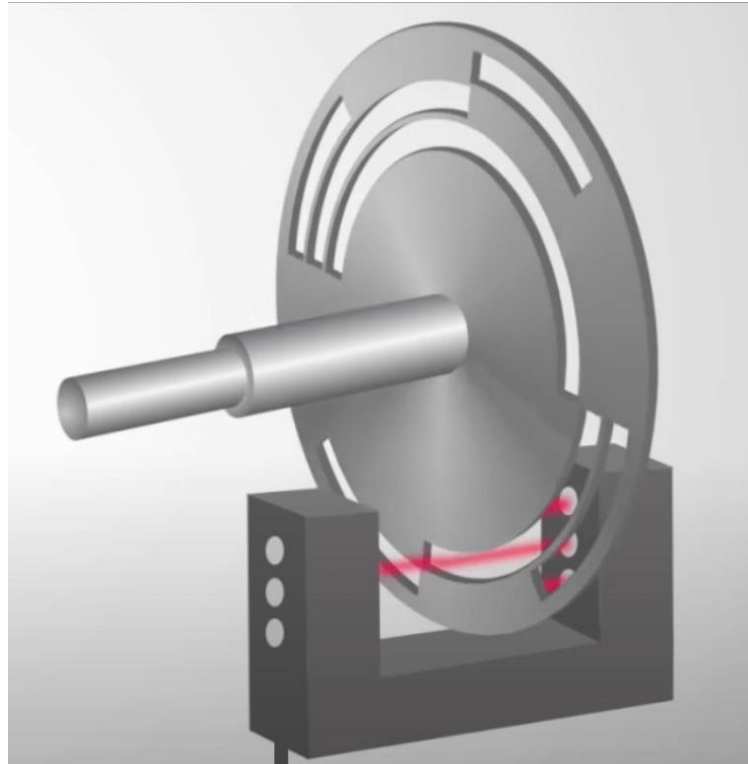
SPEED/POSITION

When the motor rotates, we have an alternation of light/dark. Higher speed, higher frequency



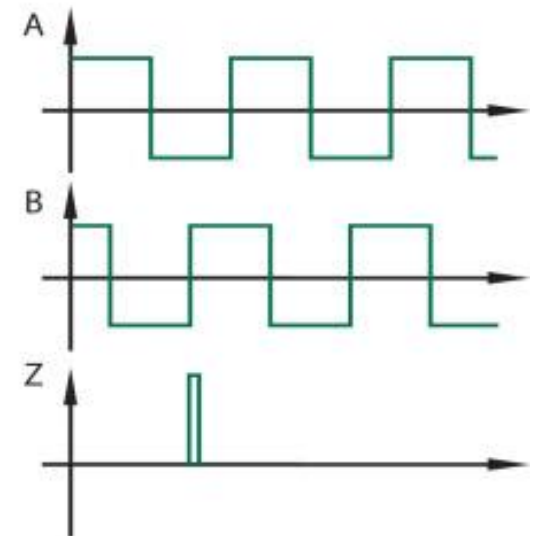
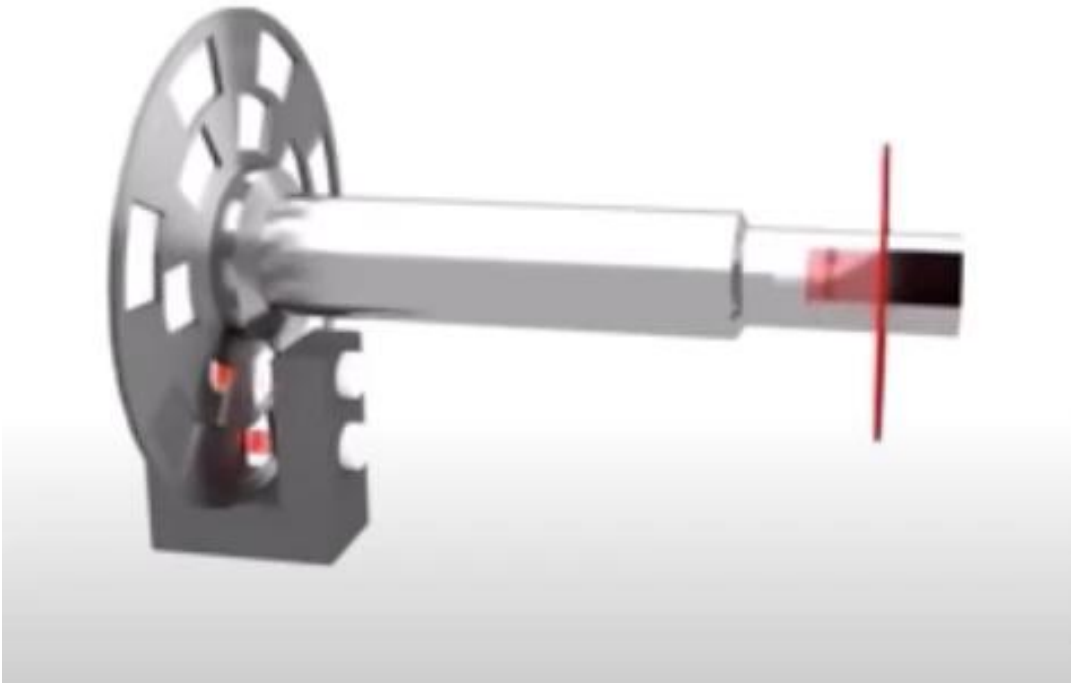
SPEED/POSITION

Based on this principle we can have absolute encoder

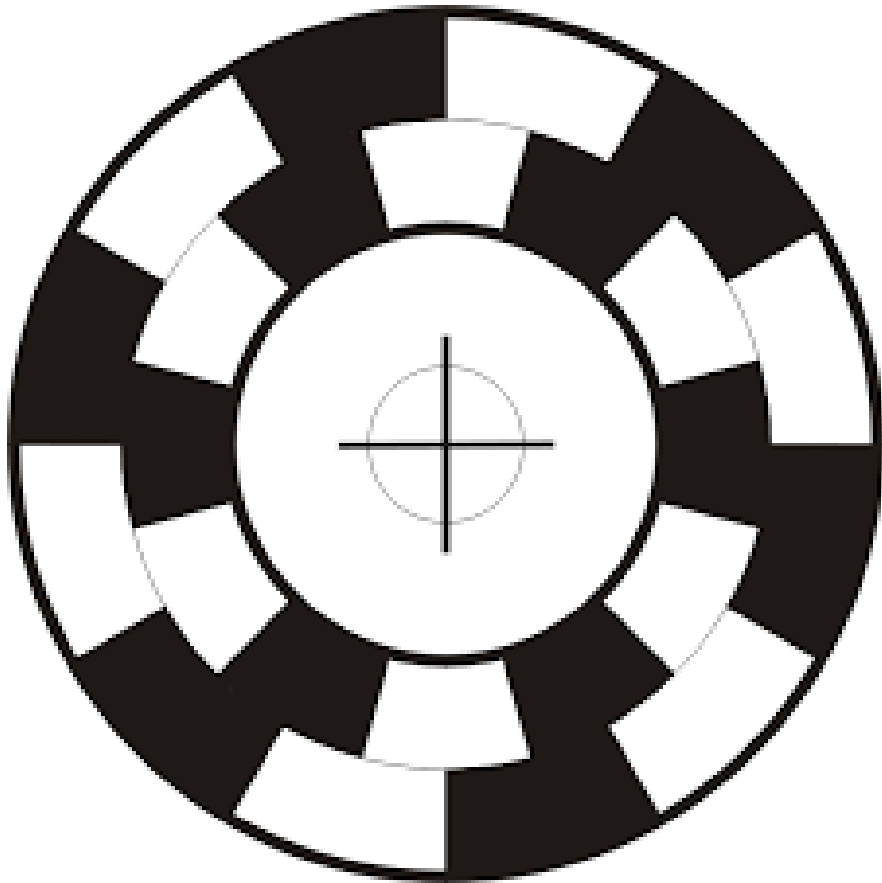


SPEED/POSITION

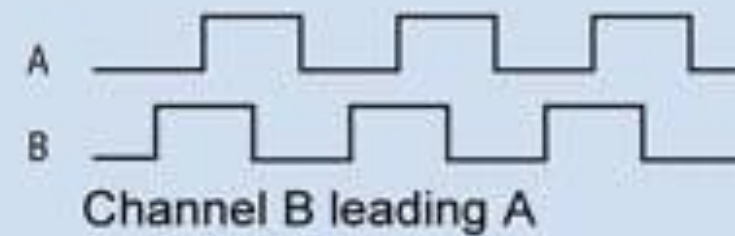
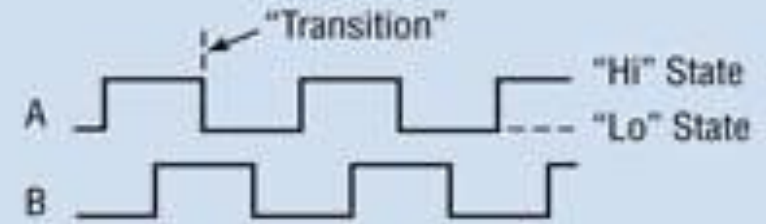
Or quadrature encoder (incremental)



SPEED/POSITION



Quadrature

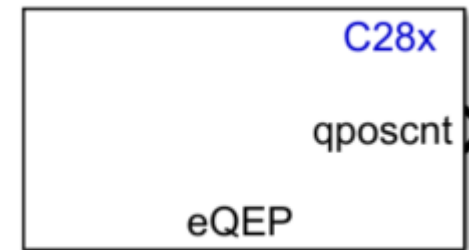
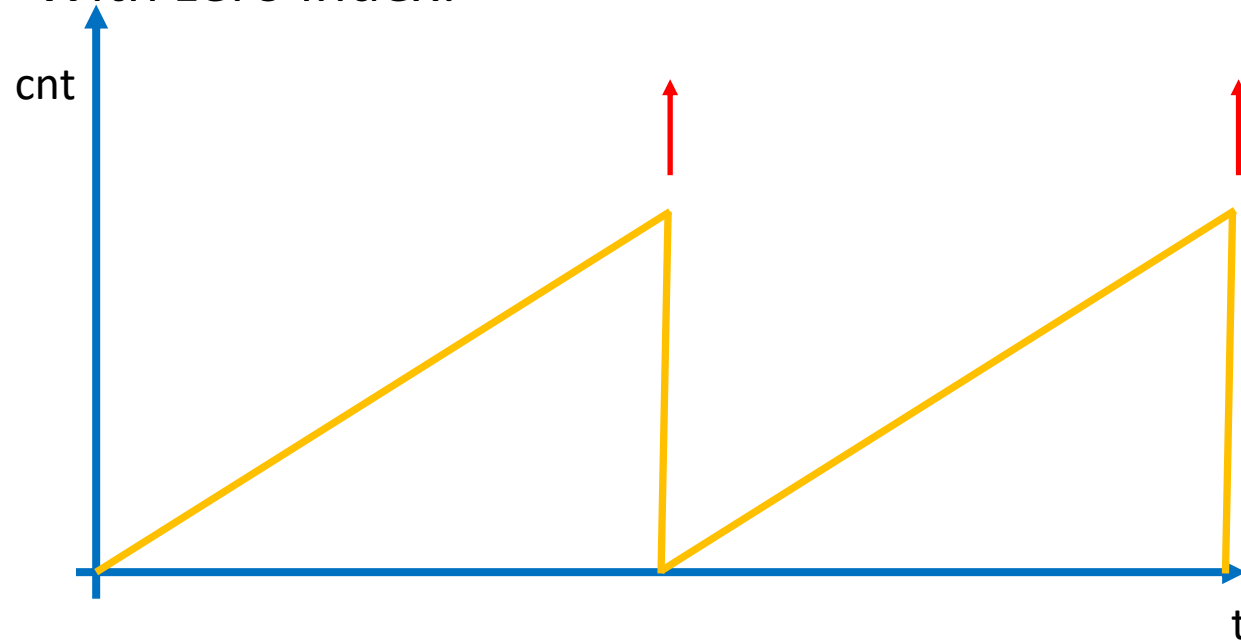


SPEED/POSITION

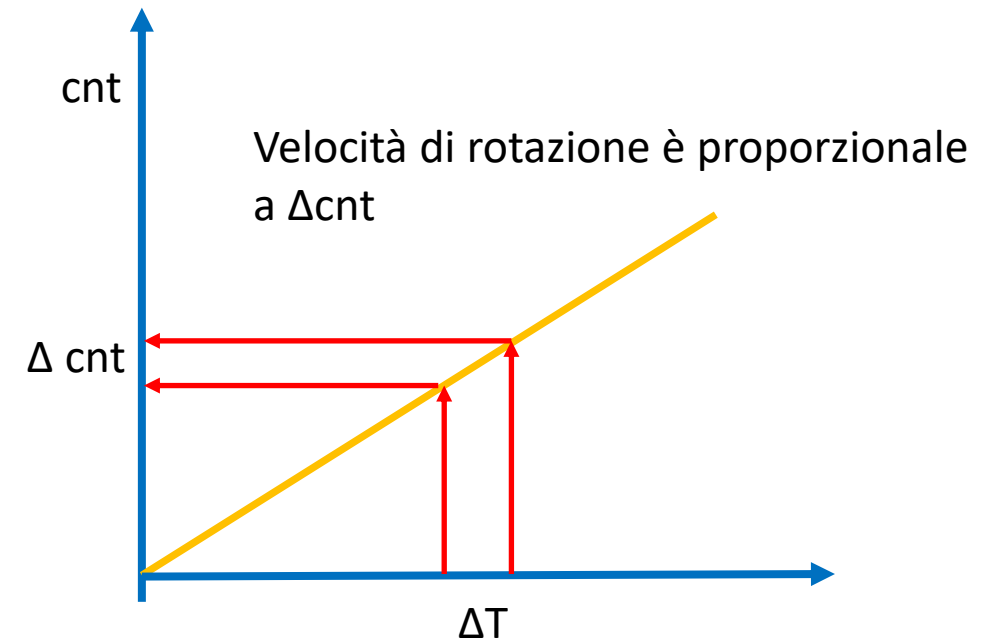
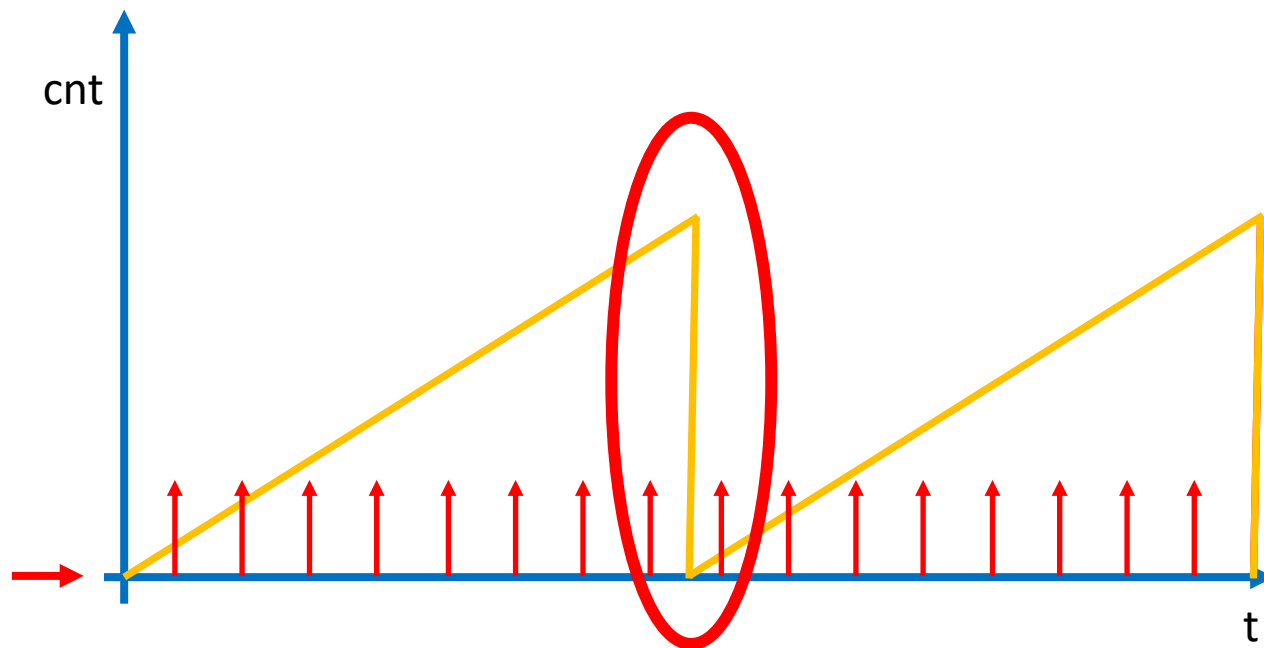
In our setup, to detect speed and position, we have a quadrature encoder with **2048 C/T**. We have 1 Rise e 1 Fall edge for Channel 1, the same for channel B.

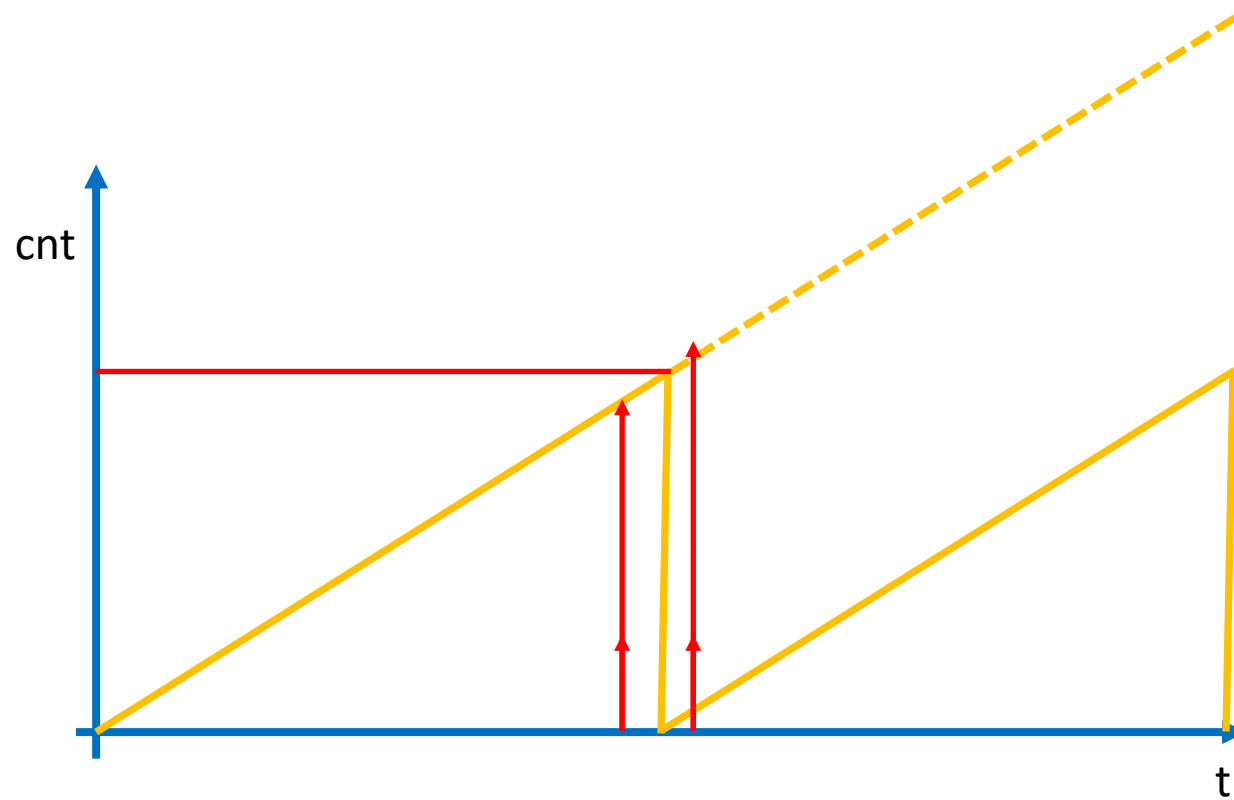
Our count per turn = **2048*4**.

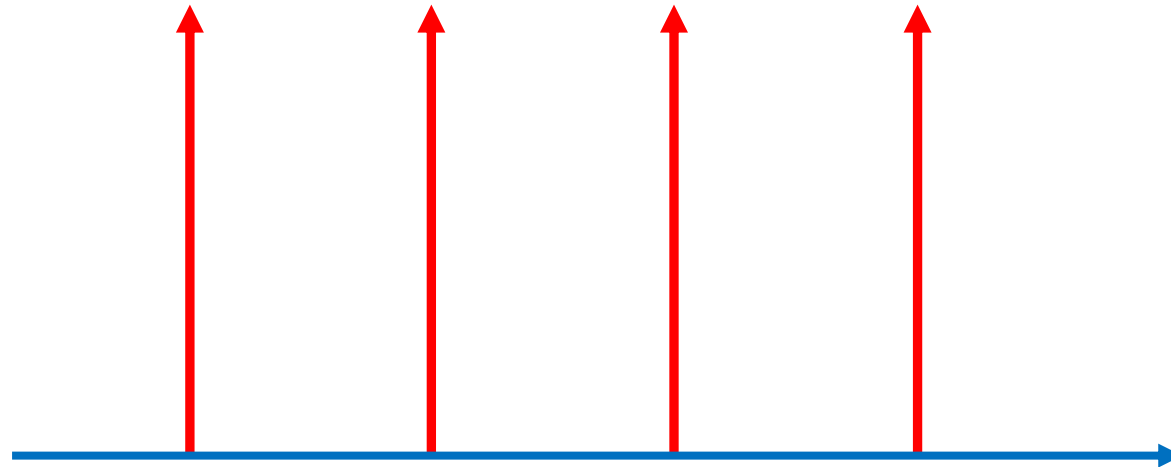
With zero index:



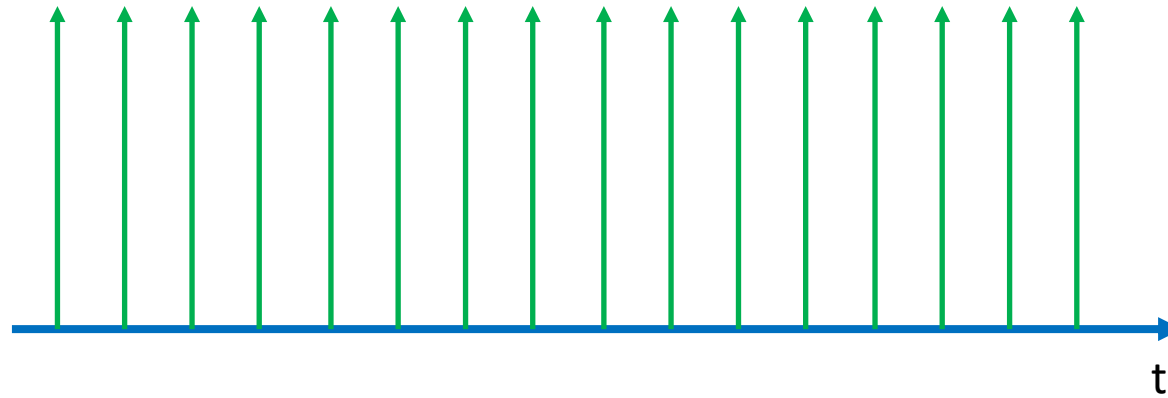
ENCODER







T_{sample}



EncoderEdge

t



In our setup:

$2048 \text{ counts/turn} * 4 \text{ edge} = 8192 \text{ counts /turn}$

If we are running at 1500 rpm we have $8192 * 1500 / 60 = 204800 \text{ counts/ sec}$

If we work at 100uS in a Sample period we have a counter increment of 20...

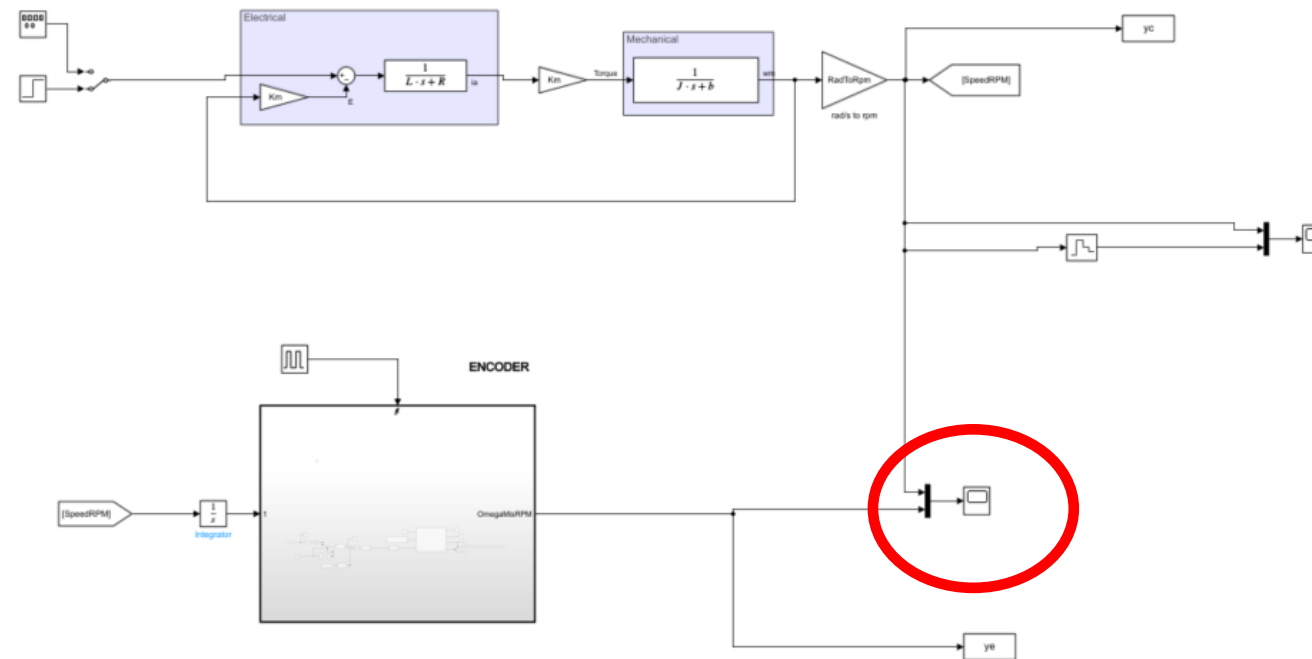
We can have an error of 1 over 20: 5% So at 1500rpm about 100rpm

We need a filter...

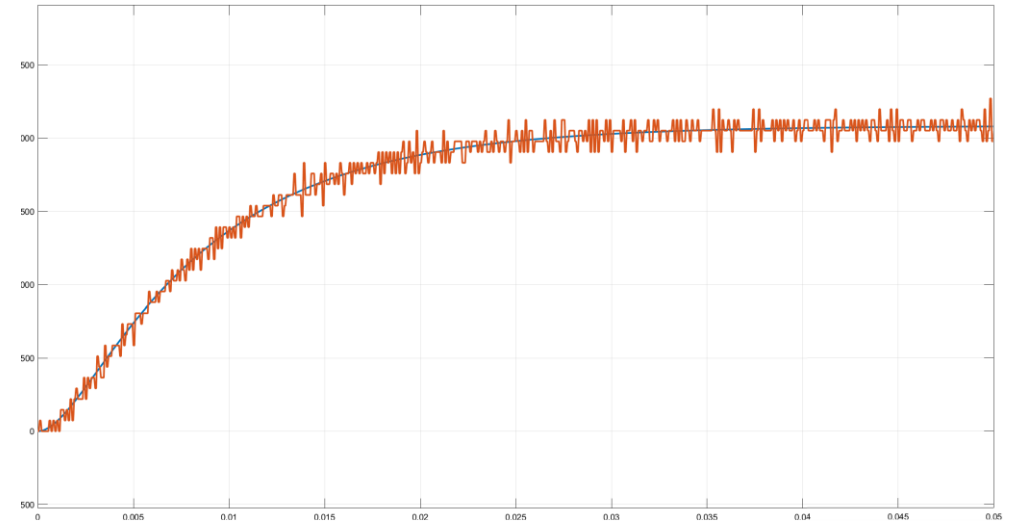
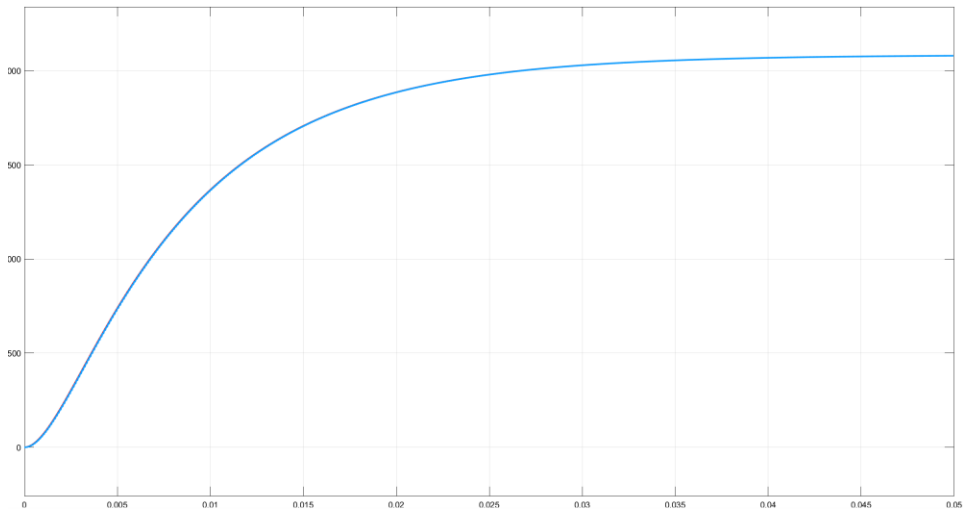
EXERCISE

The main purpose of this exercise is to show what's happen in encoder reading

Please open file **DC_MotorEquation_Encoder.slx**



The main purpose of this exercise is to show what's happen in encoder reading



Try with different Sample Time

- **0.0001** (Our Sample time)
- **0.001** (Lower value)
- **0.00001** (Higher value)

What's happen?

We need a filter, we can use

$$G(s) = \frac{1}{1 + s\tau}$$

Connect il to the encoder output

You can try different kind of filter

LOW-PASS FILTER DISCRETIZATION

$$G(s) = \frac{1}{1 + s\tau} \quad \left| \begin{array}{l} \text{backward difference method} \\ s = \frac{z - 1}{Tz} = \frac{1 - z^{-1}}{T} \end{array} \right.$$

$$G(z) = \frac{1}{1 + \frac{1 - z^{-1}}{T}\tau} = \frac{Y}{X} \quad \longrightarrow \quad Y \left(1 + \frac{\tau}{T} - \frac{\tau}{T} z^{-1} \right) = X \quad \longrightarrow \quad Y \left(1 + \frac{\tau}{T} \right) = X + Y \frac{\tau}{T} z^{-1}$$

$$Y = \frac{T}{T + \tau} X + \frac{\tau}{T + \tau} Y z^{-1}$$

$\longrightarrow$

$$Y = \frac{T}{T + \tau} X + \frac{\tau}{T + \tau} Y_{\text{prec}}$$

T=sample time

$$Y = \underbrace{\frac{T}{T+\tau}}_a X + \underbrace{\frac{\tau}{T+\tau}}_b Y_{\text{prec}}$$

Osservazioni:

- $a+b=1$
- Intuibile il filtraggio

Esempio:

$T=1$

$\tau=120$

$a=0.008264.....$

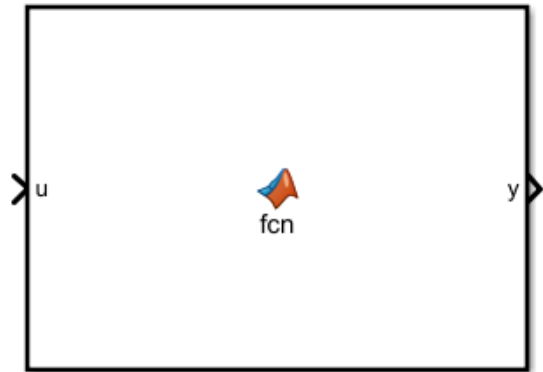
$b=0.991735.....$

$a+b=0.999$

Esempio:

$Y=0.1 X + 0.9 Y_{\text{prec}}$

Prendiamo una Matlab Function



Doppio click

```
function y = fcn(u)
```

```
y = u;
```

Prendiamo una Matlab Function



$$Y = \frac{T}{T+\tau} X + \frac{\tau}{T+\tau} Y_{\text{prec}}$$

```
function out = LowPassFilter( Ts ,Tao, In)

persistent OutPrev
if isempty(OutPrev )
    OutPrev = 0 ;
end

% Low-Pass filter
% discretized by backward difference

a=Ts/ (Ts+Tao) ;
b=Tao/ (Ts+Tao) .

out=a*In + OutPrev *b;

OutPrev = out;

end
```

Prendiamo una Matlab Function



```
function out = LowPassFilter( Ts ,Tao, In)
```

```
persistent OutPrecedente  
if isempty( OutPrecedente )  
    OutPrecedente = 0 ;  
end
```

```
% Filtro passa basso discretizzato con il  
% metodo delle differenze all'indietro
```

```
a=Ts/ (Ts+Tao) ;  
b=Tao/ (Ts+Tao) ;
```

```
out=a*In+OutPrecedente*b;
```

```
OutPrecedente = out;
```

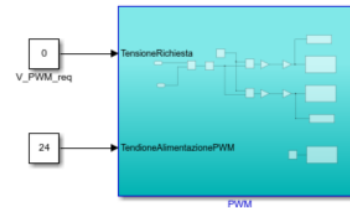
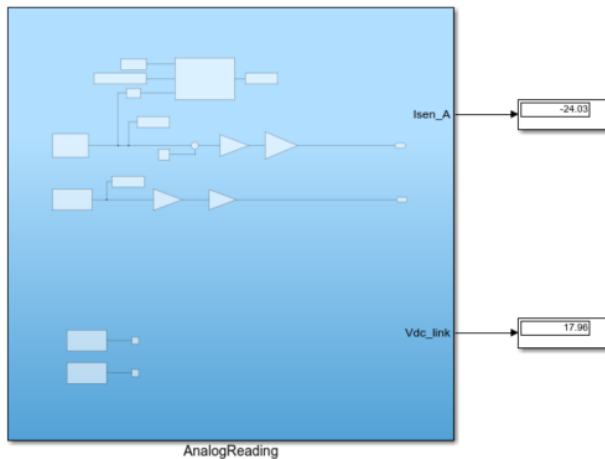
```
end
```

$$Y = \underbrace{\frac{T}{T+\tau}}_a X + \underbrace{\frac{\tau}{T+\tau}}_b Y_{\text{prec}}$$

EXERCISE

Now we can use in in our real encoder output.

Please open file **BaseModelForControl.slx**



We can run the model and check:

- Vdc_link = V Power Supply
- Isen_A
- Omega RPM

Add filter on omegaRPM

Add a filter on Isen_A

Add a filter on Vdc_link

What's about time constant?

Define it in your matlab script

